Prof. Dr. Daniel Huson

Institute for Bioinformatics
and Medical Informatics
Faculty of Science (MNF)

EBERHARD KARLS
UNIVERSITÄT
TÜBINGEN

Sequence Bioinformatics **WS 2026/27**

Assignment 3 **Due: Wed 4-Nov-2026, 10:00**

This assignment is on the lectures of Mon 26-Oct (multiple sequence alignment, second part: optimal alignment by integer linear programming, Chapter 2 of the handout, Section 2.6) and Wed 28-Oct (phylogenetics, first part: phylogenetic trees, UPGMA and Neighbor-Joining, Chapter 4, Sections 4.1 and 4.2). Exercise, section and figure numbers refer to the handout. The data of Tasks 4 and 5 are printed here, so that you need not wait for Chapter 4 to be posted.

Important: please read and follow all instructions carefully. Assignments are part of the mandatory tutorial participation. Submissions that do not follow the naming or authorship requirements will not be considered.

Setup

Download `assignment03.zip` from Ilias. It contains the skeleton `ILP4MSA_YOUR_NAME.java`, which reads the sequences and reports the counts, and two sets of three DNA sequences: `book.fasta`, the sequences of Exercise 2.15, and `long.fasta`. You also need the free solver `lp_solve` (`https://sourceforge.net/projects/lpsolve/`; on a Mac `brew install lp_solve`, on Ubuntu `apt install lp-solve`), whose input format is the one used in Section 2.6.5 of the handout. Use Java 17 or newer and no libraries: the task is to set up the program *from first principles*; the solver does the rest.

1 Three sequences by integer linear programming (4 points)

Exercise 2.15. Complete the four methods of `ILP4MSA_YOUR_NAME.java` so that

```
java ILP4MSA_YOUR_NAME <sequences.fasta> <match> <mismatch> > problem.lp
```

writes the maximum-weight-trace problem of Section 2.6.4 for the three sequences of the file as an integer linear program in the format of `lp_solve`, exactly as Section 2.6.5 does it for two sequences. The alignment edge between position i of sequence s and position j of sequence t, with $s < t$, is the binary variable `x`s`_`i`_`t`_`j (the skeleton provides the naming), weighted with `match` if the two residues are identical and with `mismatch` otherwise.

(a) The objective: one term per alignment edge.

(b) One constraint $x + x' \leq 1$ for every simple mixed cycle through two of the sequences: two different edges between the same two sequences that cross or share an endpoint (Section 2.6.5). These constraints also make every position lie on at most one edge to each other sequence.

(c) One constraint $x + x' + x'' \leq 2$ for every simple mixed cycle through all three sequences (Section 2.6.4): three edges, one between each pair of sequences, that close a cycle together with follow edges running forward in each sequence, in either of the two directions around the triangle. Three edges that meet in one column, with no follow edge in the cycle, are not a mixed cycle.

(d) The declaration of all variables as binary.

The program reports the number of variables and of constraints of each kind on standard error. **Check:** for `book.fasta` (`TCAG`, `TCCTA`, `CTAG`) with match 4 and mismatch 1 it must report 56 variables, 344 constraints through two sequences and 2840 through three, and `lp_solve -S2 problem.lp` must find the optimum 36 of Exercise 2.15. Then run `long.fasta` with the same weights: 85 variables and 9920 constraints, which `lp_solve` solves in a few seconds.

(e) For `long.fasta`, read the optimal trace off the solver's output (the variables with value 1), draw it in the extended alignment graph, write down the multiple alignment it realizes, and check its weight column by column. Is the optimal trace unique? If not, give a second trace of the same weight.

Submit your program, its counts for both files, and your answer to (e). The `.lp` files are not needed.

2 The ILP for two sequences, by hand (2 points)

Exercise 2.12: for $X = \mathtt{TCA}$ and $Y = \mathtt{TTA}$, a match score of 2 and a mismatch score of 1, write down the integer linear program of Section 2.6.5 in the form used there: the objective with its nine terms, the constraints that each position is aligned at most once, and the crossing constraints. How many constraints of each kind are there? Solve the program by inspection, and give the alignment that the optimal trace realizes and its weight.

3 Simple mixed cycles (1 point)

Exercise 2.11: let $S_1 = \mathtt{A}_1\mathtt{G}_2\mathtt{C}_3\mathtt{T}_4$, $S_2 = \mathtt{A}_5\mathtt{C}_6\mathtt{T}_7\mathtt{A}_8$ and $S_3 = \mathtt{G}_9\mathtt{A}_{10}\mathtt{C}_{11}$, the positions numbered consecutively.

(a) Draw the extended alignment graph, joining only identical residues of different sequences by alignment edges. How many alignment edges are there?

(b) List four different simple mixed cycles, two involving only two of the sequences and two involving all three, in the notation $\mathtt{A}_1 - \mathtt{A}_5 \to \mathtt{A}_8 - \mathtt{A}_1$, where "$-$" is an alignment edge and "$\to$" a path of follow edges. (This one is already taken.)

(c) For each of your cycles, state the constraint that the integer linear program of Section 2.6.4 derives from it.

4 UPGMA and Neighbor-Joining (2 points)

Exercise 4.2. Five taxa have the distances

D	a	b	c	d	e
a	0	11	9	15	19
b	11	0	14	20	24
c	9	14	0	12	16
d	15	20	12	0	10
e	19	24	16	10	0

.

(a) Run UPGMA (Section 4.2.2), giving the reduced matrix after each merge and the height of each internal node, and draw the rooted tree.

(b) Run Neighbor-Joining (Section 4.2.3): at each step give the values r, the adjusted matrix N, the pair joined and its two branch lengths, and draw the unrooted tree with its branch lengths.

(c) Which of the two trees reproduces D exactly? Check, for instance, $D(a, b)$ and $D(b, e)$ in both. UPGMA pairs a with c, although in the Neighbor-Joining tree the neighbor of a is b. Explain why, from the branch lengths of that tree and the assumption that UPGMA makes.

5 Tree distances and the Newick format (1 point)

Exercise 4.1. Take the tree `((a:9,b:8):6,c:2,(d:7,e:5):6);` of Figure 4.2a.

(a) Compute the tree distance $D_T(i, j)$ for all ten pairs of taxa.

(b) Root the tree in the middle of the edge that separates $\{a, b\}$ from the other three taxa (Figure 4.2b), and write the rooted tree in Newick format with its branch lengths. Which of the ten distances does the rooting change?